

I/O Performance and Scalability Analysis

GSoC Warm-up Task

1. Methodology & Formulas

To evaluate the scalability of the I/O pipeline, I use the following metrics:

Speedup (S): $S = T_{\text{baseline}} / T_{\text{parallel}}$. It measures how much faster the system runs with more concurrency.

Parallel Efficiency (E): $E = (S / N) \times 100\%$, where N is the number of workers/threads. It measures how effectively the added resources are utilized.

I/O Bandwidth: The rate of data transfer (MiB/s) provided by the POSIX layer in Darshan logs.

2. Local Benchmarking (GitHub Codespace)

Tested with a 5GB synthetic dataset to measure throughput and worker efficiency.

Workers	Throughput	Speedup	Efficiency	Status
1	188.45	1.00x	100.00%	Base
2	191.82	1.02x	50.89%	Drop
4	228.5	1.21x	30.31%	Peak
8	221.51	1.17x	14.69%	Saturated

Table 1: Local Benchmarking(Github Codespace)

3. HPC Characterization (LCRC Darshan Logs)

Analysis based on the POSIX module metrics from Darshan summary reports.

Threads	Run_Time	Speedup	POSIX_Bandwidth	Status
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1	244.4	1.00x	126.61	Base
2	298.64	0.82x	242.81	Regression
4	209.87	1.16x	507.56	Sweet_Spot
8	215.18	1.14x	474.52	Plateau
32	228.32	1.07x	473.5	Diminishing

Table 2: HPC Characterization(LCRC Darshan Logs)

4. Data-Driven Conclusions

The Regression at 2 Threads: The Darshan logs show that while bandwidth increased at 2 threads, the total Run Time actually increased (Speedup 0.82x). This indicates that the Metadata Overhead (file opens/seeks) and Lock Contention between threads outweighed the benefits of parallel reading.

I/O Saturation Point: The system reaches its physical I/O limit at 4 threads (~507 MiB/s). Adding more threads (8 to 32) results in a bandwidth plateau and an increase in runtime due to context-switching overhead.

Cross-Environment Consistency: Both the local Codespace (Efficiency 14.69% at 8 workers) and the LCRC cluster (Bandwidth saturation at 8+ threads) confirm that the PyTorch pipeline is I/O Bound. Increasing CPU resources (workers) cannot overcome the hardware-defined storage throughput limit.

5. Engineering Insight: Resource Constraints

During the local benchmarking, the environment hit a disk quota limit (Space < 1%). This required immediate manual intervention to manage data lifecycle. In a production HPC environment, this mimics the challenge of Storage Quota Management, where I/O performance is not just about speed, but also about capacity-aware data orchestration.

6. Attached the code execution results as your reference

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Google Sum... gemini-cli/... Create chat... Problems -... Trapping Ra... Clone Grap... Characteriz... generate_d...

Google-Summe-of-Code-warmup-io [Codespaces: humble happiness]

[Preview] README.md Extension: Python generate_data.py 1, U requirements.txt U

```
scripts > generate_data.py > generate_synthetic_data
12 def generate_synthetic_data(data_dir="..data", num_files = 10, file_size_mb = 500):
13
14
23     chunks_per_file = 5
24     chunk_size = 100 * 1024 * 1024
25
26     try:
27         with open(file_path, "wb") as f:
28             for _ in range(chunks_per_file):
29                 random_data = np.random.bytes(chunk_size)
30                 f.write(random_data)
31                 logging.info(f"Generated: {file_path}(500 MB)")
32             except IOError as e:
33                 logging.error(f"Failed to write{file_path}:{e}")
34
35     logging.info(f"Successfully generated {num_files * file_size_mb / 1024:.2f} GB of data.")
36
37 if __name__ == "__main__":
38     generate_synthetic_data()
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
python3 generate_data.py
INFO: Starting data generation in ../data...
INFO: Generated: ../data/data_chunk_0.bin(500 MB)
INFO: Generated: ../data/data_chunk_1.bin(500 MB)
INFO: Generated: ../data/data_chunk_2.bin(500 MB)
INFO: Generated: ../data/data_chunk_3.bin(500 MB)
INFO: Generated: ../data/data_chunk_4.bin(500 MB)
INFO: Generated: ../data/data_chunk_5.bin(500 MB)
INFO: Generated: ../data/data_chunk_6.bin(500 MB)
INFO: Generated: ../data/data_chunk_7.bin(500 MB)
INFO: Generated: ../data/data_chunk_8.bin(500 MB)
INFO: Generated: ../data/data_chunk_9.bin(500 MB)
INFO: Successfully generated 4.88 GB of data.
$
```

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Python - scripts

PROBLEMS 4OUTPUTDEBUG CONSOLETERMINALPORTS

Python - scripts

@JNH2 → /workspaces/Google-Summe-of-Code-warmup-io/scripts (main) \$ python3 benchmark.py

Workers | Time (s) | Throughput(S/s) | Efficiency(%)

/workspaces/Google-Summe-of-Code-warmup-io/scripts/benchmark.py:21: UserWarning: The given NumPy array is not writable, and PyTorch does not support non-writable tensors. This means writing to this tensor will result in undefined behavior. You may want to copy the array to protect its data or make it writable before converting it to a tensor. This type of warning will be suppressed for the rest of this program. (Triggered internally at /pytorch/torch/csrc/utils/tensor_numpy.cpp:213.)

return torch.from_numpy(np.frombuffer(data[:100], dtype=np.uint8))

1 | 26.53 | 188.45 | 100.00

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2 | 26.07 | 191.82 | 50.89

/home/codespace/.python/current/lib/python3.12/site-packages/torch/utils/data/dataloader.py:424: Use rWarning: This DataLoader will create 4 worker processes in total. Our suggested max number of worker in current system is 2, which is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running slow or even freeze, lower the worker number to avoid potential slowness/freeze if necessary.

self.check_worker_number_rationality()

/home/codespace/.python/current/lib/python3.12/site-packages/torch/utils/data/dataloader.py:432: Use rWarning: This DataLoader will create 4 worker processes in total. Our suggested max number of worker in current system is 2, which is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running slow or even freeze, lower the worker number to avoid potential slowness/freeze if necessary.

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4 | 21.88 | 228.50 | 30.31

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main* 0 4 0

Python

Finish Setup

Python 3.12.3

Layout: U.S.

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/home/codespace/.python/current/lib/python3.12/site-packages/torch/utils/data/dataloader.py:424: UserWarning: This DataLoader will create 8 worker processes in total. Our suggested max number of workers in current system is 2, which is smaller than what this DataLoader is going to create. Please be aware that excessive worker creation might get DataLoader running slow or even freeze, lower the worker number to avoid potential slowness/freeze if necessary.
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PROBLEMS 4

OUTPUT

DEBUG CONSOLE

TERMINAL

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Python - scripts +



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8 | 22.57 | 221.51 | 14.69
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main*



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Python



Finish Setup



Python 3.12.3



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